

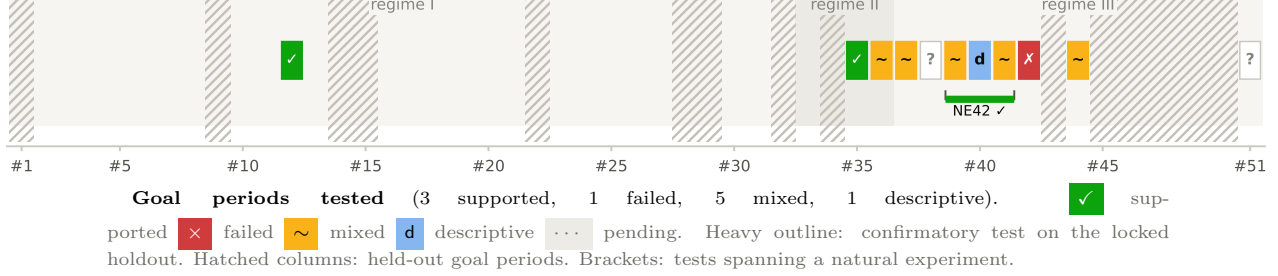
H26: The swarm is near-critical in what it says but subcritical in when it acts

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1.5/5 Usefulness 2/5

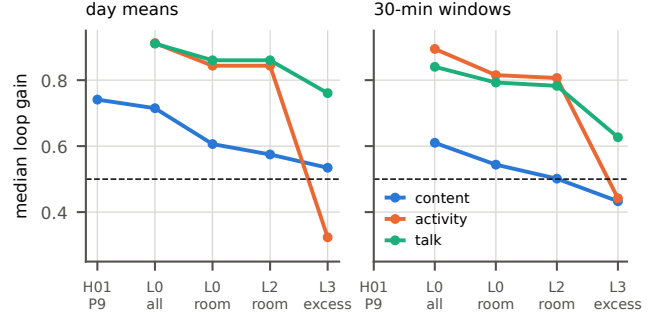
Bottom line. On corrected data, what LLM agents say and how much they talk co-move inside rooms (loop gain ≈ 0.5 , carried by the room channel) while what they do barely does; no channel is near-critical, and H01's 0.74 is mostly drives plus time-averaging.

level: hypothesis · pipeline checks 83% · updated 2026-10-04



WHAT HAPPENED

HH108 claimed the swarm's *content* is near-critical (H01: $\beta J_0/n = 0.74$) while its *activity* is subcritical (loop gains ≤ 0.4). We put content, activity and talk on one estimand, $g = 1 - 1/VR$, at matched resolution (day means, 30-min windows). Field removal was nested: dedupe, goal and kick-off directions, operator-message directions, and a within-room minus cross-room contrast. We ran 10 regime-III two-room units plus #35 and #51. Synthetic worlds at village sampling came first. Drives alone reproduce H01's number: 0.57 from a global drive at the observed cross-room level, 0.74 with more drive types. Round 1: content's room excess is genuine (median 0.53; null beaten in 6/10 units) but activity matched it (0.44 vs 0.43 at 30 min): **failed as posed**. **Round 1b** (fixed activity table, two embedding models): activity's room gain collapses (30 min 0.12; cross-room ρ 0.50 $\rightarrow$ 0.10, the event-drop artifact), so content exceeds activity (gap +0.41 [0.13, ...], 9/10 units), but talk matches content and content stays ≈ 0.5 : **unresolved**. Native NE42: the room excess is channel-borne. Holdout not run.



Observables. Median loop gain over 10 regime-III two-room units, down the field-removal ladder: H01's estimator, then whole-swarm (L0 all), room (L0), drive-projected room (L2), and room excess (L3). Content (blue) falls from 0.74 to 0.53 (day) and 0.43 (30 min). Activity and talk sit above content at every step except day-level activity L3, which is unpowered. At 30 min, activity's excess equals content's. Activity's large raw gain is village-wide co-movement.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Mean-field soft-spin $O(n)$ / Curie-Weiss (models 11, 01). Spins: an agent's 32-d whitened statement mean (content), or its fraction of active or talking minutes. Within-room coupling J ; drives: global day and time-of-day, static goal fields, measured operator inputs, unmeasured room drives. C_{ab} : cross-agent covariance of drive-removed deviations; S_a : split-half signal variance; ρ_w, ρ_c : within- and cross-room correlation; N_r : room size.

$$\rho_{w,c} = \frac{\sum_{\text{pairs}} \text{tr} C_{ab}}{\sum_{\text{pairs}} \sqrt{S_a S_b}}, \quad \rho_{\text{ex}} = \frac{\rho_w - \rho_c}{1 - \rho_c}$$

$$g = 1 - \frac{1}{1 + (N_r - 1) \rho_{\text{ex}}}, \quad g_{\text{eq}} = J, \quad g_{\text{day}} \rightarrow J(2 - J)$$

Mostly a correction, with one usable diagnostic. The HH108 gap was an artifact: it set a day-mean statistic that includes drives against a 1-min statistic with drives removed. Time-averaging alone turns an equal-time gain of 0.4 into 0.64. Compare channels at the same resolution and with the same drive removal and the gap disappears: ideas do not cascade more easily than effort. The diagnostic is *where* co-movement lives. Content co-moves within rooms (cross-room $\rho_c \approx 0.05$). Activity co-moves village-wide (ρ_c 0.5–0.8), so a swarm-wide activity dip is a platform or schedule event. Operator messages explain none of the room content co-movement. Near-critical is too strong for either channel: room loop gains of about 0.5 mean variance about twice the independent level.

H26-content-near-critical · generated 2026-10-04 10:55 by infra/summaries/build_summaries.py · card hypotheses/H26-content-near-critical/README.md · ratings: H26 agent (suggested)

RESULTS VS PREDICTIONS

Prediction (locked)	Round 1 → 1b bge / gte	Verdict
P1a drives alone give $H01 \geq 0.3$	0.57 (global); 0.67–0.74 (more drives)	passed
P2 L3 content median 0.25–0.5	0.53 → 0.52 / 0.47; ≥ 0.5 in 6 → 5 / 4 of 10	failed → met
activity L3 (day / 30 min)	0.32 / 0.44 → −0.38 / 0.12 (trimmed 0.16)	artifact
activity cross-room ρ (30 min)	0.50 → 0.10	artifact
P3 content − activity, 30 min	+0.03, 4/10 → +0.41 [0.13, ...] / +0.42, 9/10	failed → held
P3 content − activity, day	+0.32, CI $\ni 0$ → +0.76 / +0.42, CI $\ni 0$	failed
P3 content − talk, 30 min	−0.09 → −0.03 / −0.04	failed
P5 operator inputs $\leq 10\%$	≤ 0.01 in all units, both models	passed
P6 #51: content $\geq$ activity	0/4 → 1/4	failed
Card rule	failed (R2) → unresolved	changed
NE42 pseudo-rooms (native)	content, talk excess < 0 in merged #40, back at split	passed
#12 motion field (native)	on 0.72 vs off 0.66; removal halves it	passed / mixed

GOAL PERIODS IN DETAIL

G12	✓ 1b: supported (bge) / mixed (gte; N2b missed by 0.004)
G35	✓ 1b: supported
G36	~ 1b: mixed (bge) / failed (gte)
G37	~ 1b: mixed
G39	~ 1b: mixed (unchanged)
G40	d 1b: descriptive (unchanged)
G41	~ 1b: mixed (unchanged)
G42	× 1b: failed (unchanged)
G44	~ 1b: mixed (unchanged)
NE42	✓ 1b: supported (content and talk, both models; activity...)

+2 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

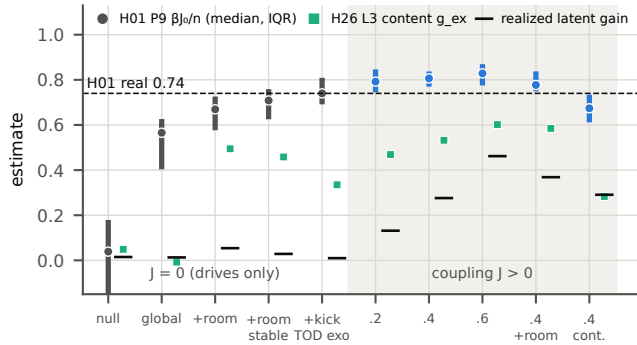
CAVEATS

Unmeasured room drives (forks in #35, instructions in #38, the override in #44) read as coupling. Day-level activity is nearly unpowered (3–8 days). Talk’s split-half variance is unreliable. The estimator favors content in synthetic worlds, which strengthens the null. One-room units lack a baseline. 1b: both embedding models agree; the trimmed activity variant keeps the 30-min gap (+0.22 [0.01, ...]). NE42 is goal-confounded. No cascade sizes (H34). The post hoc size scaling is confounded with #51.

ROUND 2

- **Essence.** whether ideas spread through a swarm more easily than effort does, measured so that channels are comparable. Round 1 says: not at matched resolution; the channels differ in where they co-move, not in how much.
- **H26-R1** Separate room coupling from room drive with the merge and split natural experiments.
- **H26-R2** Lagged, exposure-conditioned content gain.
- **H26-R3** Idea-cascade sizes (HH122) from H34.
- **H26-R4** Instrument checks: embedding swap, style control, H38 outages.

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Second figure. Synthetic validation (24 worlds each, $N = 14$ in two rooms, 5 days). With zero coupling, H01’s P9 (gray) already reads 0.57–0.74 from drives and barely moves with J . H26’s L3 (green) tracks the realized latent gain (black) under global drives. It reads unmeasured room drives (“+room”) and imperfectly measured operator pulses as coupling.